

YUKTA VAJPAYEE

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EDUCATIONAL QUALIFICATIONS

San Jose State University : Masters of Science
Computer Science

August 2025 - Present
San Jose, California

SRM Institute of Science and Technology : Bachelor of Technology
Computer Science and Engineering [CGPA: 9.58 / 10.00]

May 2022
Chennai, Tamil Nadu

TECHNICAL SKILLS AND COMPETENCIES

Languages: Java, Python, C/C++, Scala, Rust, Bash/Shell, SQL, Go, TypeScript, OpenCV, Ruby, Kotlin, C#/.NET, Swift
Technologies: Kafka, Delta Lake, Trino, PinotDB, DynamoDB, PostgreSQL, MongoDB, Docker, Kubernetes, Django, RAG Model, Hive
DevOps/Cloud: AWS (Lambda, Redshift, Athena, EMR), Apache Spark, Apache Flink, Airflow, Prometheus, Kibana, Git

WORK EXPERIENCE AND INTERNSHIPS

Hewlett Packard Enterprise – Aruba Networking
Cloud Developer - I [Data Engineer]

August 2022 – July 2025
Bangalore, Karnataka

- When temperature-driven outages were detected late, delivered a geo-aware incident service that fuses **GPS + temperature** telemetry with a rules engine to auto-open **ServiceNow** tickets, enabling proactive Problem Management and faster response.
- With limited real-time visibility across a large edge fleet, engineered a cloud-native liveness/health plane streaming status for **100K+ devices**, shrinking blind spots and triggering near-real-time remediation.
- To curb hallucinations and heavy manual analysis in Gen-AI workflows, instituted **100+** automated evals (regex + embedding similarity) and deployed **RAG** over **100K+ records**, **reducing manual query effort by 40%** and improving trustworthiness.

Hewlett Packard Enterprise – Aruba Networking
Software Development Intern

Jan 2022 – July 2022
Bangalore, Karnataka

- Under reliability/latency/cost pressure on streaming workloads, replatformed **Apache Flink** from AWS EMR to **Kubernetes**, unlocking better elasticity and operational control while improving reliability and cost efficiency.
- Modernized the analytics plane end-to-end: transitioned **Django** endpoints to resilient Python workers, tuned **Airflow/Spark** resource pools, and migrated reporting from **Athena** to **Delta Lake on S3**, improving query latency, data freshness, and overall stability.
- Built an automation service from scratch that standardized internal management workflows and scheduled actions, **saving 20+ hours of manual work per week** across teams and freeing engineers for higher-value tasks.

TATA Steel – TISCO
Data Engineering Intern

June 2020 – August 2020
Jamshedpur, Jharkhand

- Replaced batch CSV workflows with a Kafka→Flink→Delta pipeline and real-time valve dashboards, cutting detection latency from hours to seconds and enabling **60s** anomaly alerts.
- Built predictive maintenance models for critical valves using time-series features and survival analysis, improving early-warning lead time by **30%** and reducing false alarms vs threshold rules.
- Deployed bandwidth-aware edge anomaly detectors on OPC-UA gateways, cutting telemetry egress by **75%** while preserving detection precision for high-severity events.

TECHNICAL PROJECTS

ReSEA — Residency-Safe Engineering Answers | Python, FastAPI, Uvicorn, scikit-learn, Flower (FedAvg), RRF, Ollama/vLLM (opt)

- Shipped a decentralized, cross-region RAG system with **EU/US/APAC** services maintaining in-memory indexes and local click logs; runs as four local **uvicorn** processes—**no cloud required**.
- Built a coordinator that **fans out** queries to regions and **fuses** results via Reciprocal Rank Fusion (**RRF**); optional on-device LLM summarization via Ollama/vLLM.
- Added feedback-driven **federated re-ranking**: per-region logistic-regression rerankers aggregated with **Flower FedAvg**; auto-loaded from `models/reranker.joblib`.

MimicMind – Developer AI Tool | Next.js, TypeScript, TailwindCSS

- Developed MimicMind, an AI-powered full-stack platform that learns a developer's coding style and generates personalized code patches from Jira-style tickets.
- Built an interactive web workbench allowing users to upload real codebases, input feature requests, and visualize generated diffs in real time.
- Designed and deployed a modular FastAPI backend integrated with file ingestion, code analysis, and LLM-driven patch generation, hosted on Render for scalable cloud access.

Crisis - AID MatchMaker.AI | TypeScript, Prisma, Next.JS, Stripe APIs

- Developed a Crisis Aid Coordination Agent, enabling seamless donation processing and cause management through an intuitive, responsive web interface.
- Integrated Stripe APIs for secure payment processing and webhook handling, ensuring real-time updates for secure payments
- Implemented a Prisma ORM-backed database layer to manage users, causes, and donations with optimized performance and scalable schema design.